



# MERT MELİH ALBAY

albaymertmelih@gmail.com

Mustafa Kemal Mah. Çankaya/ANKARA

0 551 953 58 72

[Personal Website/ Github Portfolio](#)

## ABOUT ME

Senior Electrical & Electronics Engineering student with over two years of practical experience in hardware architecture, embedded systems, and real-time data fusion. Specialized in developing AI-driven autonomous systems with the proven ability to lead projects independently from conceptual design to functional prototyping.

## EDUCATION

Mugla Sıtkı Kocman University  
Electrical Electronics Engineering (English)| 2022-2026 (Expected)

## EXPERIENCES

Kalyon Enerji Yatırımları A.Ş. | Intern | 07/07/2025 - 04/08/2025  
Executed site layout design for solar PV plants in AutoCAD, conducting production efficiency and shading analysis using PVsyst and MATLAB. Leveraged Python to process real-time SCADA data from wind turbines, engineering mathematical models for power forecasting and performance evaluation.

Chemnitz University of Technology | Chemnitz, Germany | Aug 2026 (Expected)  
Invited by Prof. Dr. Florian Röhrbein for a one-month academic research stay, I will actively participate in research activities and scientific exchange focused on neurorobotics and intelligent systems. During this period, I aim to gain hands-on insight into ongoing laboratory projects, specifically bridging the gap between hardware integration and intelligent control systems.

## LANGUAGE

ENGLISH - C1

## SKILLS

MATLAB | Altium Designer | Python | VHDL | OpenCV | AutoCAD | PVsyst | SolidWorks  
Embedded Systems | RF Signal Processing | Elektromechanical Systems  
Computer Vision | Image Processing | Sensor Fusion

## PROJECTS

Deep Learning-Based Optical Inspection and Visual Defect Detection Module

CNC-Probe: 3-Axis Robot for Thermal and Electrical PCB Analysis

High-SNR Amplifier for Low-Amplitude Signals

Analysis of Synchronous Generator Performance Under Dynamic Loading Using a MATLAB-Based GUI

## LICENSES & CERTIFICATIONS

General Electric Aerospace- Explore Engineering Job Simulation | 2024

MIT EDX Course- Circuits And Electronics: Circuit Analysis I | 2025

Sivil Havacılık Genel Müdürlüğü- Sportif/Amaör IHA Lisansı

Dijital Donanım Tasarımcılığı Kursu (FPGA / ASIC) | 2026